

Supplementary Material:

Exact Radial-Determinant Frequencies of a Piezoelectric Annular Plate, and Inverse Recovery of e_{31} from Open-Circuit Flexure

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DRAFT — September 24, 2026

Main text contains no job numbers; they are listed in §S.6.

S.1 Cubic consistency condition

Each layer carries $\varphi = \pm \bar{\varphi}f(s)$, $s = |z| - h \in [0, h_1]$, with $f(0) = f(h_1) = 0$ (mirrored so that the two layers' moments add). The thickness-integrated electric enthalpy of one layer is $-\bar{e}_{31}I_1\bar{\varphi}\nabla^2w - \frac{1}{2}\bar{\Xi}_{33}I_2\bar{\varphi}^2 - \frac{1}{2}\bar{\Xi}_{11}I_3|\nabla\bar{\varphi}|^2$ plus the elastic part, with $I_1 = \int zf' dz = -\int f ds$, $I_2 = \int f'^2 ds$, $I_3 = \int f^2 ds$. Stationarity in w and $\bar{\varphi}$ gives the flexural equation with piezo moment coefficient $K = -2\bar{e}_{31}I_1$ (two layers) and the electrical equation $\bar{\Xi}_{11}I_3\Delta\bar{\varphi} - \bar{\Xi}_{33}I_2\bar{\varphi} - \bar{e}_{31}I_1\Delta w = 0$; the insulated edge $\partial\bar{\varphi}/\partial r = 0$ is its natural condition. The long-wave increment per layer is $(\bar{e}_{31}I_1)^2/(\bar{\Xi}_{33}I_2) \leq h_1^3\bar{e}_{31}^2/(12\bar{\Xi}_{33})$ by Cauchy–Schwarz, with equality only for f' linear in s , i.e. the quadratic bubble $f = 4s(h_1 - s)/h_1^2$ of the main text: $I_1 = -2h_1/3$, $I_2 = 16/(3h_1)$, $I_3 = 8h_1/15$, so $K = \frac{4}{3}h_1\bar{e}_{31}$ and the normalized electrical equation is $a\Delta\bar{\varphi} - \bar{\varphi} + c\Delta w = 0$ with $a = h_1^2\bar{\Xi}_{11}/(10\bar{\Xi}_{33})$, $c = h_1^2\bar{e}_{31}/(8\bar{\Xi}_{33})$. The sine of Duan, Quek and Wang [1] gives $K = (4/\pi)h_1\bar{e}_{31}$; under this projection its long-wave increment is $96/\pi^4$ of exact, and under their thickness-integrated Gauss-law closure ($a = h_1^2\bar{\Xi}_{11}/(\pi^2\bar{\Xi}_{33})$, $c = h_1^2\bar{e}_{31}/(2\pi\bar{\Xi}_{33})$) it is $12/\pi^2$ of exact. The earlier version of this paper used that closure; the solver keeps it as a switch (`projection='duan'`), and it reproduces every earlier number bit-for-bit. Positing $\bar{\varphi} = \chi\bar{w}$ with χ independent of r gives $\lambda = \chi/(a\chi + c)$ from the electrical equation and $(d_1 + d_2)\lambda^2 + K\chi\lambda - A_2\omega^2 = 0$ from the flexural one; eliminating λ gives the cubic $Ka\chi^3 + (d_1 + d_2 + Kc - A_2\omega^2a^2)\chi^2 - 2A_2\omega^2ac\chi - A_2\omega^2c^2 = 0$ (Duan, Quek and Wang's Eq. 17a is the same cubic up to a constant factor, with their coefficients). Each of the three roots (χ_i, λ_i) depends on the trial frequency ω . For each root the radial problem is $\Delta\bar{w} - \lambda_i\bar{w} = 0$, with Bessel pair (J_n, Y_n) or modified pair (I_n, K_n) according to the sign of λ_i . The general short-circuit solution is the sum over all three branches. The limit $\bar{e}_{31} \rightarrow 0$ is singular (leading and two subleading coefficients of the cubic vanish together), so the elastic bilayer is a separate 4×4 .

The open-circuit rank-1 coefficient of the main text,

$$\alpha = \frac{4\bar{e}_{31}^2(h + h_1/2)^2h_1}{\bar{\Xi}_{33}(r_o^2 - r_i^2)}, \quad (\text{S.1})$$

is the unique sign that raises the first free-free $n = 0$ root. It is the product of the moment arm of M_p and the charge arm of Q , both $h + h_1/2$ (the bus charge is $-\partial\mathcal{H}/\partial V$ of the same enthalpy); the earlier outer-face charge arm $h + h_1$ made α too large by $(h + h_1)/(h + h_1/2)$. On a mixed edge the same αS is added only where $M_{rr} = 0$ is imposed (the free edge). Driven Q/V uses the same 4×4 with right-hand side $M_p = 2\bar{e}_{31}(h + h_1/2)V$ on both moment rows. An independent bisection of this Q/V recovers the open-circuit root to 9×10^{-15} relative.

S.2 Inverse on-branch cross-check

For the two clamped-clamped short-circuit recoveries, the product $(d\omega/dp)(dp/d\omega)$ on the solution branch is 0.999999 (e_{31}) and 1.000008 (E). Propagated uncertainties at 0.10% frequency precision are ten times the 0.01% values in main-text Table 4: 0.246% of E , 3303% of short-circuit e_{31} , and 8.70% of open-circuit e_{31} at $h_1/2h = 1/12$.

The sequential recovery (main text §6) was run at full production precision (dps=100). Round trip: $\hat{E} = 2.000000009 \times 10^{11}$ Pa (4.5×10^{-9} relative), then, holding $\hat{E}$, $\hat{e}_{31} = -4.099999346$ C m⁻² (1.6×10^{-7} relative); both below the 10^{-6} bar. On-branch sensitivities at $h_1/2h = 1/12$: $d\omega_{SC}/dE = 5.678 \times 10^{-9}$ rad s⁻¹ Pa⁻¹, $d\omega_{OC}/de_{31} = -2.1375$ rad s⁻¹ m² C⁻¹, $d\omega_{OC}/dE = 1.494 \times 10^{-9}$ rad s⁻¹ Pa⁻¹, giving an implicit $de_{31}/dE|_{\omega_{OC}} = 6.992 \times 10^{-10}$ C m⁻² Pa⁻¹. At 0.01% frequency precision this splits the 1.21% total of main-text Table 4 into a 0.870% direct term and a 0.839% term from propagating $\hat{E}$'s own 0.0246% uncertainty, added in quadrature; at 0.10% frequency precision the same split gives 8.70% and 8.39%, total 12.08%. The recovered $\hat{e}_{31}$ is on the printed branch, not the reflection twin at 22.0 C m⁻². The negative control substitutes the clamped-clamped open-circuit root for the free-free one in the second step: $\hat{e}_{31} = -4.09975379$ C m⁻² still solves (sign flip present), but $d\omega_{SC}/de_{31}|_{\hat{E}} = -2.062 \times 10^{-2}$ rad s⁻¹ m² C⁻¹ propagates to 330.3% of e_{31} at 0.01% frequency precision – the single-parameter clamped-clamped figure of main-text Table 4, confirming the control fails as required.

S.3 Finite-element discriminator and mixed electrical boundary

Axisymmetric PLANE223, KEYOPT(1)=1001, millimetre-newton-second-tonne, relative permissivity, one mesh for every electrical condition compared. Mode 1 is the numerical rigid body (≤ 0.004 Hz). Mode 2 is bending: transverse displacement same sign and magnitude at the host top and bottom at $r = r_o$; radial displacement opposite. The ratio of those components at $h_1/2h = 1/12$ is 19.71 (both-faces short-circuit) and 20.06 (open-circuit). Open-circuit mechanical amplitudes are smaller under unity-normalization because VOLT participates in the norm; the ratio is the identifier. Every number in this section is from the $e_{31} = -4.1$ C m⁻² rerun of the decks (job 2532086, §S.6); coupling-off decks carry no e_{ij} and were not rerun.

The outer-only-ground deck (VOLT= 0 on $z = \pm(h + h_1)$, interface free) gives 121.859 Hz. That is 2.56% above true both-faces short-circuit (118.816 Hz), and 0.60% above open-circuit, and is not used as the short-circuit baseline in the main text. Coupling-off is 118.810 Hz.

At $h_1/2h = 1/8$: mode 2 is 124.722 Hz open-circuit, 121.364 Hz short-circuit, discriminator ratio 20.20 and 19.70. At $1/5$: mode 2 is 132.183 Hz open-circuit, 127.105 Hz short-circuit, discriminator ratio 20.43 and 19.70. All runs show mode 1 at 0 Hz (rigid body), end with RUN COMPLETED, and report zero in the MAPDL NUMBER OF ERROR MESSAGES ENCOUNTERED line. (The original +4.1 decks carried the literal string *** ERROR *** in a pre-registration comment, which a queue script's text scan once misreported as a failure; the rerun decks neutralize that token.)

The same pair with the piezo through-thickness division count doubled from 2 to 4, leaving radial divisions, host through-thickness divisions, units, and electrical BC unchanged, moves the three splits by at most 4×10^{-5} of their value (1.9438%, 2.7661%, 3.9951% against 1.9438%, 2.7662%, 3.9953%). The finite-element splits are 99.7%, 99.6% and 99.5% of the tabulated closed-form Δ_{OC} , and 99.7%, 99.7% and 99.7% of the interior-enriched open-circuit root over the short-circuit 6×6 (1.9493%, 2.7747%, 4.0080%), which is the like-for-like quantity.

Sign control. The same decks were first run with $e_{31} = +4.1 \text{ C m}^{-2}$ ($\bar{e}_{31} = -4.85$; jobs 2491761, 2491831, 2494226, 2494665, 2495593 of §S.6). Their splits, 0.2795%, 0.4047% and 0.6002% (F–F) and 0.2593% (C–F), are 99.9%, 99.9%, 99.9% and 98.9% of the closed form at that sign. Across the two signs the finite-element F–F split grows by factors of 6.95, 6.84 and 6.66 at the three ratios, against closed-form factors of 6.97, 6.85 and 6.68; the open-circuit update tracks $\bar{e}_{31}^2$, not one calibrated point. An earlier version of this model took the bus charge at the outer face (arm $h + h_1$); against it the +4.1 finite elements recovered 93%/90%/86%, which is the charge-arm ratio $(h + h_1/2)/(h + h_1) = 0.929/0.900/0.857$ and not a mesh effect.

S.4 Wide-axis admittance

A harmonic PLANE223 run on the same mesh (job 2532146; 1 V on the outer-bus master, KEYOPT(1)=1001 charge reaction, 118.55–122 Hz in 70 substeps) places the $|Y|$ peak at 118.80 Hz, the charge changing sign through the pole between 118.80 and 118.85 Hz (118.82 Hz by linear interpolation of $1/Q$), and the charge-zero at 121.126 Hz, against the modal pair 118.816 / 121.126 Hz. The same deck in job 2532086 swept only the narrower window of the $e_{31} = +4.1$ run and did not reach the zero; it is superseded. At +4.1 (job 2494863) the same construction placed the pair at 118.79 / 119.13 Hz against 118.811 / 119.143 Hz. An earlier harmonic dump (job 2494669) used the current label AMP5, which is KEYOPT(1)=101, and returned a numerical-zero bus current at every frequency; that run is not a disproof of the pole/zero.

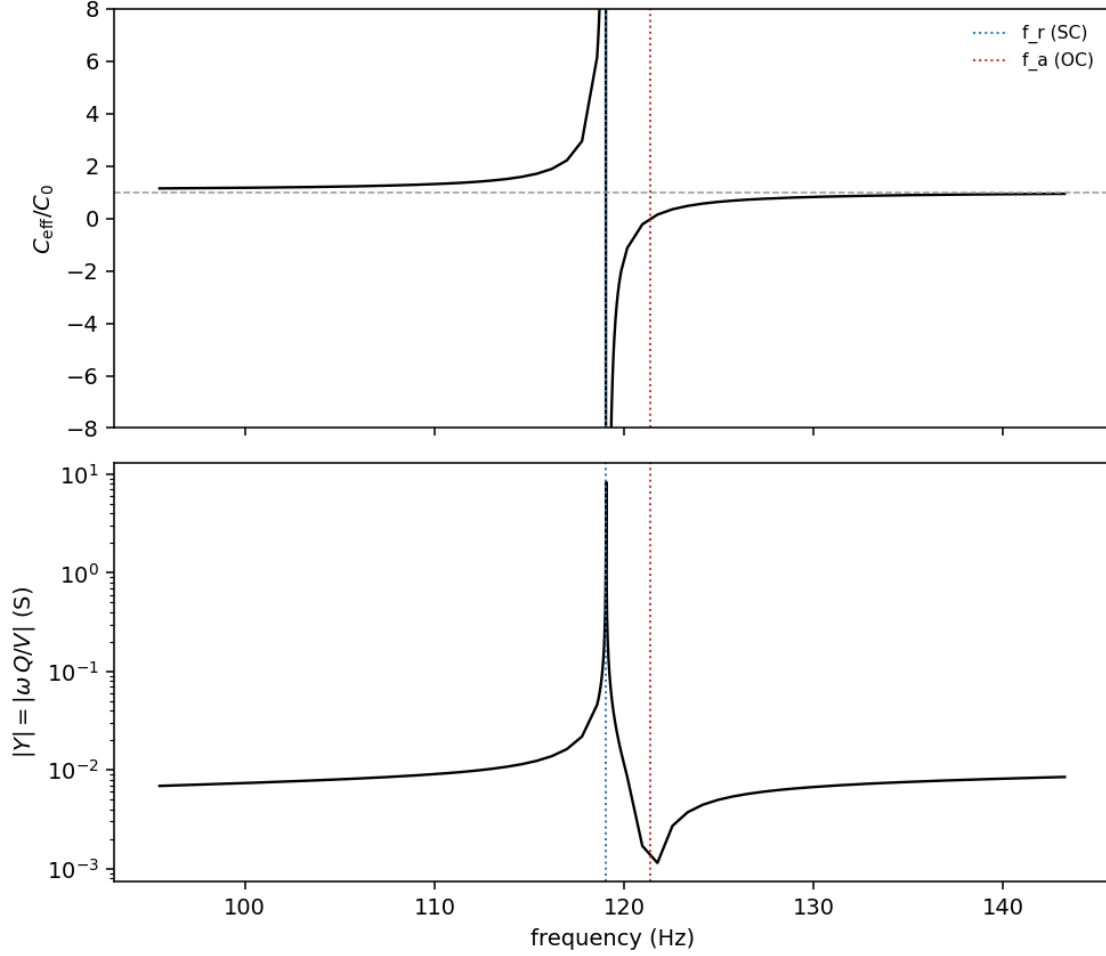


Figure S.1: Driven C_{eff}/C_0 and $|Y|$ on 95–145 Hz at $h_1/2h = 1/12$. The pole and zero of the first $n = 0$ pair differ by 1.95%; main-text Fig. 4 zooms 116–122.5 Hz.

S.5 Thickness-ratio sweep and other-ratio admittance

Figure S.2 is the closed-form $h_1/2h$ sweep that sits underneath main-text Table 2 and Fig. 2: eleven ratios from $1/40$ to $1/4$, linear-potential open-circuit 4×4 against the elastic bilayer. The three Duan Table 4 ratios (red) lie on the curve. The thin-layer intercept $\Delta_{\text{OC}}/(h_1/2h) \rightarrow 0.25$ is the leading h_1 scaling of the rank-1 coefficient α (the same limit under either charge arm, since $h_1 \rightarrow 0$). The bottom panel is the single-parameter e_{31} G2 at 0.01% frequency precision (host E known exactly); it is 2.60% at $1/40$ and 0.40% at $1/4$.

Figure S.3 is the driven Q/V problem at the other two Table 4 ratios. Independent bisection of Q/V recovers `oc_ff_bisect` to 1.7×10^{-14} (1/8) and 3×10^{-16} (1/5); k_{eff}^2 from the driven pole/zero is 0.0533 and 0.0757. $C_0 = 6.66 \mu\text{F}$ at $1/8$ and $4.16 \mu\text{F}$ at $1/5$. Off-resonance Butterworth–van Dyke shape: $C_{\text{eff}}/C_0 = 1.135$ at $0.55 \omega_{\text{el}}$ and 0.797 at $1.08 \omega_{\text{OC}}$ at $1/8$; 1.201 and 0.740 at $1/5$.

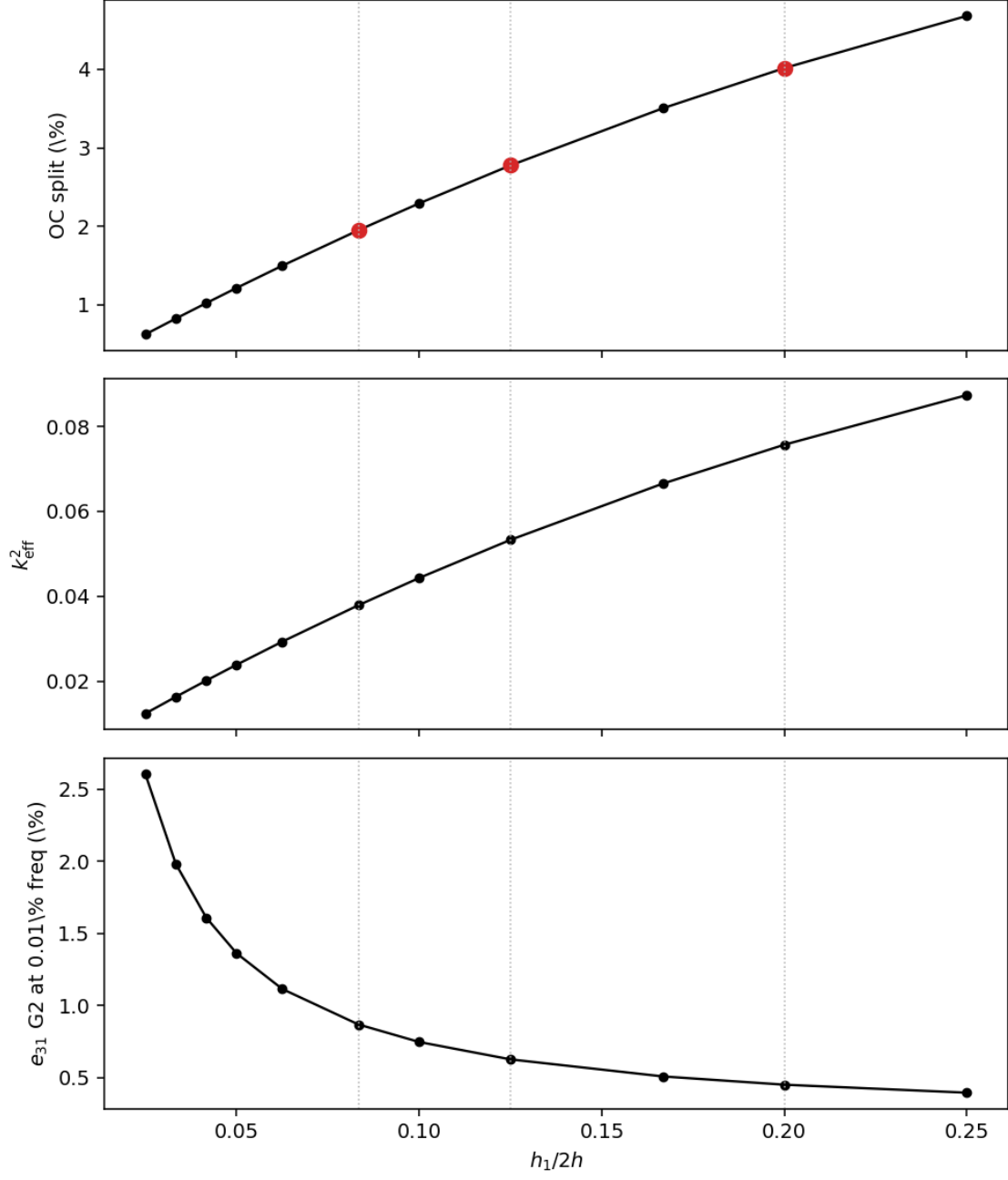


Figure S.2: Closed-form F-F $n = 0$ open-circuit split, k_{eff}^2 , and e_{31} G2 versus $h_1/2h$. Red markers are the three Duan Table 4 ratios already in the main text.

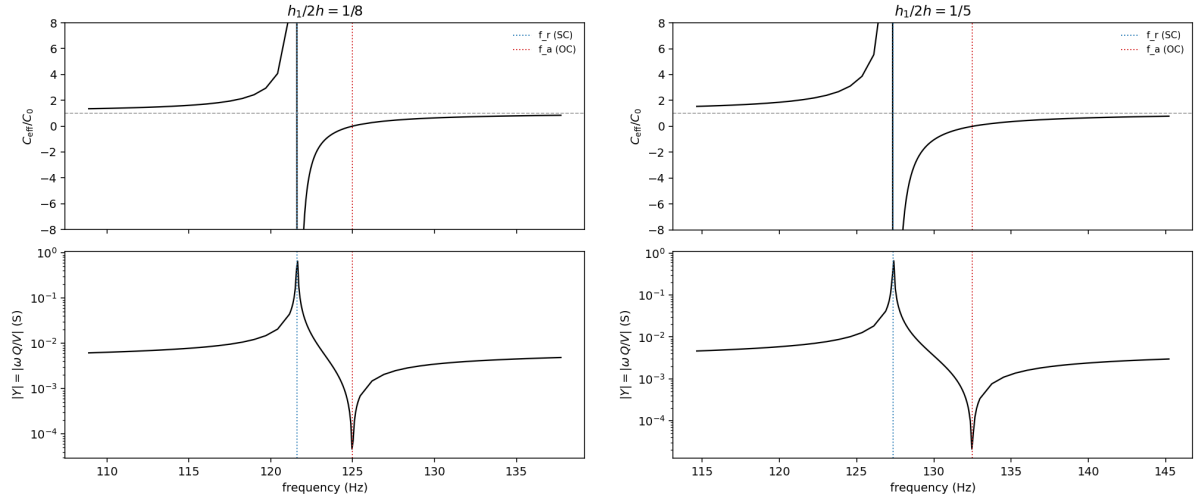


Figure S.3: Driven C_{eff}/C_0 and $|Y|$ at $h_1/2h = 1/8$ (left) and $1/5$ (right). Pole at the elastic/short-circuit root, zero at the open-circuit root, same linear-potential operator as main-text Fig. 4.

S.6 Cluster job provenance

All jobs on The Mill [5], solver snapshot 2026-07-10.s10. Piezoelectric paths are new and unused by the elastic default. Jobs up to 2531012 used $e_{31} = +4.1 \text{ C m}^{-2}$; the paper's numbers are from the $e_{31} = -4.1$ rerun described at the end of this section, and the earlier jobs are kept as the sign control of §S.3.

- 2489140 — clamped-clamped Table 4 gate, 27/27, worst -0.0476% .
- 2490706 — F-F coupling-off and outer-only-ground decks at $h_1/2h = 1/12$.
- 2490848 — F-F outer-only-ground at $h_1/2h = 1/8$ and $1/5$.
- 2491531 — clamped-clamped short-circuit inverse of E and e_{31} .
- 2491661 — F-F open-circuit python archival, deploy miss (package lacked the open-circuit method); superseded by 2491764.
- 2491761 — F-F open-circuit finite elements, mode 2 = 119.143 Hz.
- 2491764 — F-F open-circuit inverse archival, all gates passed.
- 2491831 — both-faces short-circuit finite elements, mode 2 = 118.811 Hz.
- 2491920 — driven $Y(\omega)$ archival, all gates passed, $k_{\text{eff}}^2 = 0.005997133$.
- 2494216 — sequential E -then- e_{31} inverse archival at full **dps**=100, $h_1/2h = 1/12$; SENTINEL PASS_ALL, bit-identical to the in-sandbox validation run.
- 2494226 — F-F open-circuit and both-faces short-circuit finite elements at $h_1/2h = 1/8$ and $1/5$, cloned from 2491761/2491831.
- 2494665 — lever 8 mesh refinement, NDIV_ZP $2 \rightarrow 4$ on the fair OC/scboth pair at all three Table 4 ratios; splits unchanged.
- 2494667 — lever 2 $h_1/2h$ sweep archival, SENTINEL PASS_ALL, bit-identical to the in-sandbox run.
- 2494668 — lever 5 driven $Y(\omega)$ at $1/8$ and $1/5$ archival, SENTINEL PASS_ALL, bit-identical to the in-sandbox run.
- 2494669 — lever 9 first harmonic dump; MAPDL clean but **RF,AMPS** identically zero. Superseded by 2494863.
- 2494863 — lever 9 v2 harmonic Y at $1/12$, **RF,CHRG**; pole 118.79 Hz, zero 119.13 Hz.
- 2494864 — lever 3 other-constant inverses, SENTINEL PASS_ALL.
- 2494865 — lever 1 $n = 0$ overtones; one overtone in $(800, 4000) \text{ rad s}^{-1}$ ($+0.137\%$ split). Pre-registered bar of two overtones in that window not met.
- 2494945 — lever 1 wide scan $(800, 15000) \text{ rad s}^{-1}$, SENTINEL PASS_ALL; second overtone 8515.06 vs $8516.48 \text{ rad s}^{-1}$ ($+0.017\%$).

- 2494866 — lever 4 F–F E channel, SENTINEL PASS_ALL, 0.025% of E at 0.01% freq on both elastic and OC.
- 2495528 — lever 10 C–F / F–C mixed-edge archival, SENTINEL PASS_ALL, bit-identical to the in-sandbox `dps=60` run.
- 2495593 — optional C–F FE, clones of 2491761/2491831 with inner $U_X = U_Y = 0$ only. scboth 67.071 Hz, OC 67.245 Hz, +0.259% (98.9% of the consistent closed-form +0.262%; 92% of the Gauss-law +0.282%). Both RUN COMPLETED, 0 MAPDL errors.
- 2530336 — first submission of the three-dimensional $n \geq 1$ queue (§S.9); no deck ran (the cluster had retired the ANSYS 2024 R2 module that every earlier finite-element job used).
- 2530423 — three-dimensional $n \geq 1$ queue, ten SOLID226 decks, ANSYS 2025 R1, all RUN COMPLETED, 0 MAPDL errors, 0 warnings (§S.9).
- 2530553 — thin-limit first pass, five bare-steel SOLID186 decks ($t = 20, 10, 5$ mm), ANSYS 2025 R1, all RUN COMPLETED, 0 errors, 0 warnings (§S.9).
- 2530671 — thin-limit second pass, first submission; the $96 \times 288 \times 8$ decks exceeded the 64 GB memory request (in-core Block Lanczos, 67.8 GB) and were killed; the added $t = 10$ mm coarse deck completed.
- 2530784 — thin-limit second pass, $96 \times 288 \times 8$ at $t = 20, 10, 5$ mm, 100 GB; all RUN COMPLETED, 0 errors (§S.9).
- 2530987 — Liu solid-disk decks, first version (§S.10); both RUN COMPLETED, 0 errors. A displacement constraint on the central mesh hole blocked the $n = 1$ centre rotation and raised those modes by 2–7%; superseded by 2531012 and not used.
- 2531012 — Liu solid-disk decks with the hole free, SOLID226, ANSYS 2025 R1; both RUN COMPLETED, 0 errors (§S.10).
- 2495751 — lever 16 frequency-precision sweep, SENTINEL PASS_ALL, bit-identical to the in-sandbox run (F–F/C–F e_{31} uncertainty curve plus the Boeringa–McCune lab floor).

Consistent-projection rerun (2026-09-23). Every analytic job above (all but the finite-element jobs) used the Gauss-law closure of §S.1 and the outer-face charge arm. Every package-based probe was re-run against the corrected solver (default `projection='consistent'`; `projection='duan'` reproduces the jobs above bit-for-bit), with only the hard-coded open-circuit regression anchors of four probes moved to the new values. All passed.

$e_{31} = -4.1$ **rerun (2026-09-24).** The piezoelectric material bundle now carries $e_{31} = -4.1 \text{ C m}^{-2}$ as printed in Liu, Wang and Quek’s and Duan, Quek and Wang’s Table 1. Every package-based probe behind a number in the main text or this document was copied with $e_{31} = -4.1$ and every e_{31} bracket moved to $[-10, -1] \text{ C m}^{-2}$ (which excludes the reflection twin at 22.0). Magnitude windows that were pre-registered for $e_{31} = +4.1$ were multiplied by $(13.05/4.85)^2 = 7.24$ before the run. Regression anchors that encoded +4.1 values were re-anchored to the -4.1 value of an independent probe, never to the probe’s own output: the thickness-ratio sweep to the open-circuit and driven- Q/V probes, the driven Q/V probes to the thickness sweep’s free-vibration roots, and the precision sweep to the open-circuit and mixed-edge probes, and the wide overtone scan to the narrow one. Eleven of the twelve gated probes end SENTINEL PASS_ALL. The twelfth, the narrow

$n = 0$ overtone scan, fails its pre-registered bar of two overtones below 4000 rad s^{-1} exactly as job 2494865 did at +4.1 (there is one), and the wide scan finds the second; four short scripts supply Table 4 in both closures, the clamped-clamped short-circuit e_{31} sensitivity, the interior-enriched open-circuit root, and the solid-disk roots of §S.10. The runs were made at the probes' production precision (dps 40–100) on a two-core cloud workstation, not on the cluster; every earlier cluster archival run of these probes reproduced its off-cluster run bit-for-bit. The package test suite passes (110 tests; four of them pin the -4.1 default, the Table 4 fundamentals, the open-circuit split and the monolithic short-circuit root, and the solid-disk anchors of §S.10 were moved to their -4.1 values). Finite elements: job 2532086 (ANSYS 2025 R1) reran every coupled deck of §S.3, §S.7, §S.9 and §S.10 at $e_{31} = -4.1$, all RUN COMPLETED with zero MAPDL errors; job 2532146 reran the harmonic admittance deck on a window that contains the new zero. Decks, text outputs, manifests, scorers and queue logs are in `ansys/NewAnsys/` of the public code tree.

S.7 Mixed-edge C–F / F–C closed form

Paper 3 names inner then outer: C–F is clamped inner, free outer (`bc_inner='C', bc_outer='F'` in `ring_disk.py`). Methods on `PiezoOutOfPlaneSolver`: `elastic_cf_det / elastic_fc_det`, `cf_coupled_det / fc_coupled_det`, `oc_cf_det / oc_fc_det`. The mixed-edge engine recovers the existing free–free and clamped–clamped determinants bit-exactly when both edges are set equal. Bare-host $n = 0$ roots match Paper 3's live Frobenius solver at relative 4.45×10^{-10} (C–F, $410.074181 \text{ rad s}^{-1}$) and 2.28×10^{-10} (F–C, $873.221632 \text{ rad s}^{-1}$). SENTINEL PASS_ALL (`probe_piezo_p4_cf_mixed_edge_2026-09-16.py`, `dps=60`; $e_{31} = -4.1$ copy `probe_piezo_p4_cf_mixed_edge_e31m_2026-09-24.py`): C–F open-circuit +1.826%, $k_{\text{eff}}^2 = 0.0355$, e_{31} G2 0.93% at 0.01% frequency; F–C open-circuit +0.0388%, $k_{\text{eff}}^2 = 0.00078$. Short-circuit 6×6 over the mixed bilayer is +0.0061% (C–F) and +0.0039% (F–C). Inverse round-trip on C–F e_{31} is 4.0×10^{-9} . C–F finite elements (job 2532086): `scboth` 67.076 Hz, OC 68.290 Hz, split +1.811% (99.2% of +1.826%), no 0 Hz rigid body, bending discriminator 45.5 and 47.0, 0 MAPDL errors. The +4.1 control (job 2495593) gave +0.259% against +0.262%. SOLVER_VERSION remains 2026-07-10.s10.

S.8 Frequency-precision sweep and lab noise floor

Table S.1 extends the two-column (0.01%, 0.1%) e_{31} inverse of the main-text open-circuit section and §S.7 to an eight-point log-spaced grid of assumed relative frequency error, on both the free–free and clamped–free $n = 0$ open-circuit channels, using the identical linear-propagation formula (unchanged, not retuned): $\delta e_{31}/e_{31} = (\delta\omega/\omega)\omega_{\text{target}}/(|d\omega/de_{31}|e_{31,\text{true}})$. The lab-floor row (marked) is sourced from Table 3 of [4]: an HP4294A impedance analyzer and a scanning laser Doppler vibrometer measured the same five axial resonances of a free–free PZT-4 rod specimen (a physically distinct specimen from this ring, same material system), disagreeing by 0.12%, 0.10%, 0.09%, 0.11%, and 0.09% – mean 0.102%. The companion draft (`BoeringaMcCune2026-2.pdf`) was checked and rejected as a source: its impedance-versus-SLDV table is internally inconsistent (disagreements up to 26% on a single mode) and carries explicit placeholder text, so no number was taken from it.

Delivered as `Paper4_Piezo/probe_piezo_p4_lever16_freq_noise_floor_2026-09-16.py` (LF) with matching submit script (`dps=60`, mirroring the free–free/clamped–free probes' own settings); its $e_{31} = -4.1$ copy (`..._e31m_2026-09-24.py`) produced the table above, SENTINEL

Table S.1: Propagated relative uncertainty in e_{31} versus assumed relative frequency error, $h_1/2h = 1/12$. G0 round-trip error is 4.0×10^{-9} (bar 10^{-6}) at every row; G_anchor reproduces the 0.01% entries of main-text Table 4, and the 0.10%-scaled figures of §S.2, to within 1% relative; the linearity self-check (every row equal to the base row scaled by the ratio of assumed frequency errors) is exact to 10^{-9} .

assumed rel. freq. error	F-F OC unc. of e_{31}	C-F OC unc. of e_{31}
0.001%	0.087%	0.093%
0.003%	0.261%	0.278%
0.01%	0.870%	0.928%
0.03%	2.610%	2.785%
0.1%	8.701%	9.282%
0.102% (Boeringa-McCune floor [4])	8.875%	9.468%
0.3%	26.104%	27.847%
1%	87.013%	92.824%

PASS_ALL. The +4.1 version was archived on the cluster (job 2495751) bit-identical to its off-cluster run. SOLVER_VERSION remains 2026-07-10.s10.

S.9 Three-dimensional $n \geq 1$ finite-element check

SOLID226 (KEYOPT(1)=1001) for host and skins, three glued volumes, host with zero piezoelectric constants and relative permittivity 1, skins PZT-4 poled $+z$; millimetre-newton-second-tonne, relative permittivity. Half ring $0 \leq \theta \leq \pi$, $U_Y = 0$ on $y = 0$: the $\cos n\theta$ family at every n , three rigid modes left. Electrical cases as in §S.3: coupling-off ($e_{ij} = 0$), both-faces short-circuit, and open-circuit (interfaces grounded, one coupled VOLT set on both outer faces). Meshes $24 \times 72 \times (2+1+1)$ and $40 \times 144 \times (4+2+2)$ (radial, circumferential, host and each skin). Circumferential order is read from a discrete cosine transform of U_Z at 13 angles on five radii; every scored mode has purity 1.0000. Coarse-to-fine changes are at most 0.025% in frequency and 0.2% in the short-circuit shift. ANSYS 2025 R1 (job 2530423 at $e_{31} = +4.1$; coupled decks rerun at -4.1 in job 2532086); the axisymmetric coupling-off deck ran under 2024 R2, and the $n = 0$ coupling-off root agrees with it to 0.0003%. Every quantity below is a ratio of raw MAPDL frequencies. Criteria were fixed before the run.

Table S.2: Three-dimensional finite elements (fine mesh) against the closed form, first radial root. Shift: short-circuit over coupling-off minus one (finite elements) and short-circuit 6×6 over elastic 4×4 minus one (closed form). Offset: coupling-off over the elastic bilayer minus one. Open-circuit minus short-circuit is below 3×10^{-10} relative at every $n \geq 1$.

$h_1/2h, n$	FE shift	closed-form shift	FE/closed -1	FE coupling-off (Hz)	offset
1/12, 0	5.230×10^{-5}	5.212×10^{-5}	+0.3%	118.8105	-0.198%
1/12, 1	4.929×10^{-5}	4.920×10^{-5}	+0.2%	276.6199	-1.022%
1/12, 2	1.045×10^{-5}	1.048×10^{-5}	-0.3%	72.3441	-0.533%
1/12, 3	1.367×10^{-5}	1.396×10^{-5}	-2.1%	172.2871	-0.647%
1/12, 4	1.706×10^{-5}	1.742×10^{-5}	-2.1%	301.9394	-0.935%
1/5, 0	5.300×10^{-4}	5.285×10^{-4}	+0.3%	127.0378	-0.262%
1/5, 1	4.984×10^{-4}	4.978×10^{-4}	+0.1%	295.4942	-1.192%
1/5, 2	1.059×10^{-4}	1.065×10^{-4}	-0.5%	77.4742	-0.604%
1/5, 3	1.382×10^{-4}	1.417×10^{-4}	-2.5%	184.4153	-0.753%
1/5, 4	1.722×10^{-4}	1.768×10^{-4}	-2.6%	322.9683	-1.097%

Against the pre-registered criteria: the axisymmetric control, mesh convergence, the open-circuit/short-circuit coincidence at $n \geq 1$, and the shift ($\pm 10\%$, all ten within $\pm 2.6\%$; within $\pm 1.2\%$ in the +4.1 run, where the shift is 7.24 times smaller) pass. The elastic offset criterion, $|\text{offset}| \leq 1\%$ at $n = 1-4$, fails at three of eight points (1/12 $n = 1$; 1/5 $n = 1$ and 4). It was set on the assumption that the offset would grow from its $n = 0$ value with frequency alone; the $n \geq 1$ points instead carry an extra -0.2% to -0.46% beyond a plane-wave transverse-shear and rotary-inertia estimate, which accounts for the $n = 0$ offset. The retired Q_r free-edge row is 10.5% off the finite elements at 1/12, $n = 1$.

Thin-limit follow-up. To separate thickness physics from an n -dependent error in the thin-plate operator, the same half ring was run as bare steel ($E = 200$ GPa, $\nu = 0.3$, no skins) at full thickness $t = 20, 10$ and 5 mm (SOLID186, full integration). Kirchhoff roots scale exactly with t , so any offset that survives $t \rightarrow 0$ is an operator error. Each thickness was run on three uniformly refined meshes, $24 \times 72 \times 2$, $48 \times 144 \times 4$, and $96 \times 288 \times 8$ (the last 2.93×10^6 equations), and each root was Richardson-extrapolated with its observed order p . Table S.3 gives the mesh-extrapolated $\delta_n(t) = f_{\text{FE}}/f_{\text{K}} - 1$ and the intercept c of the quadratic $\delta = c + at + bt^2$ through the three thicknesses.

Table S.3: Bare steel annulus, first radial root, Richardson-extrapolated mesh limit. c_∞ : quadratic extrapolation to $t = 0$; c_x : the same from the finest mesh without Richardson; at , bt^2 at $t = 20$ mm; p : observed convergence order at $t = 5$ mm.

n	$t = 20$	$t = 10$	$t = 5$	c_∞	c_x	at	bt^2	p
0	-0.158%	-0.040%	-0.010%	-0.000%	+0.000%	-0.000%	-0.158%	3.3
1	-0.918%	-0.361%	-0.161%	-0.014%	+0.013%	-0.481%	-0.422%	1.5
2	-0.490%	-0.223%	-0.111%	-0.013%	+0.009%	-0.363%	-0.114%	1.5
3	-0.583%	-0.236%	-0.106%	-0.005%	+0.009%	-0.346%	-0.231%	1.9
4	-0.836%	-0.320%	-0.137%	-0.004%	+0.012%	-0.434%	-0.398%	2.1

Criteria were fixed before each run. A first pass on the two coarser meshes was formally unresolved: its coarse-to-fine change at $t = 5$ mm exceeded the 0.05% bar at three orders, so its intercept (then +0.004% to +0.055%) was not scored. The second pass added the finest mesh and required monotone ratio-2 convergence with $1 \leq p \leq 8$ at every thickness and order (met at all fifteen points), took the intercept from the Richardson limits in that case, and required $|c| \leq 0.03\%$ at $n = 0-4$: all five pass, with $|c_\infty| \leq 0.014\%$. A separate criterion, fine-to-finest change at most 0.02%, fails at two points ($t = 5$, $n = 1$ and 4: 0.023% and 0.024%) and does not enter the intercept by the pre-registered rule. Convergence is monotone from above, so the finest-mesh intercept c_x and the Richardson intercept c_∞ bracket the limit; both are within 0.015% of zero at every order. An operator error of the size of the $n \geq 1$ excess in the main text would have given c of -0.2% to -0.46% . The $n = 0$ offset is quadratic in t to three figures (ratios 3.99 and 3.97 per halving). At $n \geq 1$ the fitted linear term is as large as or larger than the quadratic one, the signature of a free-edge boundary layer rather than interior shear alone. The finest $t = 5$ deck printed one Block Lanczos ill-conditioning warning (test residual 3.3×10^{-3} at the -1 Hz shift); its roots continue the monotone sequence of the two coarser meshes, and no other deck printed a warning or an error.

S.10 Liu 2002 solid-disk short-circuit CPT

Liu, Wang and Quek [2] tabulated free-vibration frequencies of a steel circular plate (no hole) with identical thickness-polarized PZT-4 skins on both faces, both faces of each skin short-circuited. Only

their classical-plate column x_{kir} is scored here; their Mindlin and three-dimensional finite-element columns are not. Geometry of their Tables 2 and 6: $r_0 = 0.6$ m, host half-thickness $h = 0.01$ m, skin thickness $h_1 = 0.002$ m ($r_0/h = 60$); materials as their Table 1, including $e_{31} = -4.1$ C m $^{-2}$, as in the main text.

The solid disk has its own determinant. Keeping only the centre-regular Bessel function of each of the three χ -branches of Section S.1 gives a 3×3 ; its rows are $w = w' = 0$ (outer edge clamped, Liu’s Tables 2–5) or $w = M_r = 0$ (simply supported, Tables 6–9), each with the short-circuit row $\phi' = 0$. The annular 6×6 evaluated at $r_i = 0$ is not this determinant, because its Y_n and K_n columns are singular there, and the annular solver refuses $r_i \leq 0$. With $h_1 = 0$ the disk reduces to the 2×2 on $\{J_n, I_n\}$ of McCune and Stutts [3], and it reproduces their λ^2 for $n = 0, 1, 2$, $m = 1, 2$, under both edge conditions, to 5×10^{-6} . Roots were located by sampling the determinant on a uniform grid (800 points from 50 rad s $^{-1}$ to 8000 (clamped) or 7000 (simply supported)) and bisecting every sign change at 100 digits; no published value placed a bracket. The first two roots of each n are the six modes of each table, with no extra or missing root in the window.

Tables S.4 and S.5 give the result. All twelve roots agree with x_{kir} to better than 1×10^{-4} , with both signs of error, and eleven of the twelve round to Liu’s printed value (the twelfth, 6193.976 against 6193.9, is 1.2×10^{-5} off). With $e_{31} = +4.1$ the same roots all fell below x_{kir} , by up to 9×10^{-5} , beyond the rounding of the higher modes.

This comparison tests the electromechanical coupling only weakly. On this geometry, changing e_{31} from -4.1 to 0.01 C m $^{-2}$ (so $\bar{e}_{31}^2$ falls from 170 to 80 C 2 m $^{-4}$) moves every root by $4\text{--}6 \times 10^{-5}$, and replacing the projection of Section S.1 by the Duan-type closure of [1] moves it by 2×10^{-5} , both comparable to the rounding of the printed values. What the tables confirm is the solid-disk determinant, both outer edge rows, and the laminate stiffness and inertia on a geometry other than the ring of the main text. Script and output: `validation/paper4_piezo/probe_piezo_p4_liu_disk_gate2_2026-09-24.py` (run at the -4.1 default).

A three-dimensional finite-element model of the same disk (SOLID226 piezoelectric bricks, both faces of each skin grounded, half model, one mesh of 6912 elements with a free central hole of radius 3 mm, 0.5% of r_0) reproduces Liu’s ABAQUS column for all twelve modes within the pre-registered 0.6%: from +0.22 to +0.52% for the clamped edge and from -0.01 to +0.31% for the simply supported one. Like theirs, it lies below the classical-plate roots, by 0.02 to 2.0% growing with mode order, which is the thickness effect at $r_0/h = 60$. This checks the finite-element pipeline used elsewhere in this paper against an independent code; it is not a test of the classical-plate roots (job 2532086 at $e_{31} = -4.1$; job 2531012 at $+4.1$ differs from it by at most 0.007%; decks and scorers in `ansys/NewAnsys/ansys_p4_liudisk2_*` and `score_liudisk2_fe_e31m_2026-09-24.txt`).

Table S.4: Liu Table 2 (outer edge clamped, short-circuit), $r_0/h = 60$: our roots ω against Liu’s classical-plate column x_{kir} (rad/s).

n	m	ω (this work)	x_{kir} (Liu)	$\omega/x_{\text{kir}} - 1$
0	1	902.4785	902.5	-2.38×10^{-5}
1	1	1878.1694	1878.2	-1.63×10^{-5}
2	1	3081.0798	3081.1	-6.56×10^{-6}
0	2	3513.4317	3513.4	$+9.01 \times 10^{-6}$
1	2	5373.6789	5373.7	-3.92×10^{-6}
2	2	7472.1342	7472.1	$+4.58 \times 10^{-6}$

Table S.5: Liu Table 6 (outer edge simply supported, short-circuit), $r_0/h = 60$: our roots ω against Liu's classical-plate column x_{kir} (rad/s).

n	m	ω (this work)	x_{kir} (Liu)	$\omega/x_{\text{kir}} - 1$
0	1	435.6424	435.6	$+9.73 \times 10^{-5}$
1	1	1227.4965	1227.5	-2.82×10^{-6}
2	1	2262.4418	2262.4	$+1.85 \times 10^{-5}$
0	2	2625.2406	2625.2	$+1.54 \times 10^{-5}$
1	2	4282.4347	4282.4	$+8.11 \times 10^{-6}$
2	2	6193.9762	6193.9	$+1.23 \times 10^{-5}$

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